SOFTWARE CAPABILITIES REFERENCE — REVISION 20

Complete reference for the MCO-1D flight software layer. Covers all ground station commands, the learning and optimisation architecture, all 29 hypotheses, all 8 mission foci, all knowledge files, and the tick execution order. Current codebase: 57,834 lines, 96/96 audit checks passing as of August 22, 2026.

Section	Topic
1	Tick Execution Order — what runs every 20ms and in what sequence
2	Complete GS Command Reference — all commands, grouped by subsystem
3	29 Aerodynamic Hypotheses — full list with targets and V ranges
4	8 Mission Foci — scorers, aliases, and seeds
5	Learning Architecture — discovery → MAP-Elites → strategy genome
6	Knowledge Files — what persists and where
7	Pilot Authority Constants — PA doctrine parameters
8	GPS Degradation Reference — 5 levels and thresholds
9	Surface Classifier Reference — evidence sources and thresholds
10	WIG Floor Speed-Curve — floor height by airspeed

SECTION 1 TICK EXECUTION ORDER

Every tick at 50Hz — in sequence

The tick wrapper (_tick_with_range_check) runs every 20ms. Order is critical — each system sees the output of all systems before it.

#	System	Function	Notes
1	Core tick	_orig_fc_tick(self, sensors)	Original MCOFlightController.tick — per-wing kinematics
2	S1 thermal	thermal.update()	Real FET+motor temps from ODrive CAN
3	H-33 morphing	h33_tick	Variable wing area — only when active
4	H-34 morphing	h34_tick	Dynamic sweep — only when active
5	H-35 morphing	h35_tick	Thrust/lift separation — only when active
6	H-36 morphing	h36_tick	Loiter/ferry mode — only when active
7	Morphing exploration	morphing_exploration_tick	Trial lifecycle for morphing discovery sessions
8	S1 health	s1_monitor.update()	Per-wing fault detection
9	Hover speed gate	hover_mode_update_speed	Updates hover/FW blend from pitot
10	Pilot intent	pilot_intent_tick	Roll/pitch/yaw/lever → 15 mechanisms
11	Hover control	hover_control_tick	V<3m/s hover: lever→alt rate, pedals→yaw
12	Mechanism learning	_mechanism_learning_tick	Weight nudges + surface classifier @1Hz + sg eval
13	Airborne range check	mco_airborne_range_check	One-time in-flight range test
14	CG management	cg_management_tick	CG shift tracking and alerts
15	Progressive failure	progressive_failure_tick	Wing health monitoring, auto land-now
16	Asym failure trim	asymmetric_failure_attitude_tick	Attitude bias for asymmetric thrust
17	Discovery	discovery_tick	Hill-climber + hypotheses + focus + MAP-Elites
18	Dart control	dart_control_tick	Dragonfly lateral+yaw model + flap_glide_tick
19	Attitude hold	attitude_hold_tick	Electronic dihedral, sets PA reference targets
20	Gust response	gust_response_tick	pa_gust_compensation + storm_resilience_tick
21	Conversion corridor	conversion_corridor_tick	Guards morphing commands during transition
22	Position hold	position_hold_tick	GPS position lock + transit_tick
23	Turn coordination	turn_coordination_tick	Yaw bias for banked turns
24	Airspeed hold	airspeed_hold_tick	Frequency trim for constant speed
25	Adv mode optimizer	advanced_mode_optimizer_tick	Aggregates mode metrics
26	WIG floor	wig_floor_tick	LAST — speed-aware minimum altitude, overrides all

SECTION 2 GS COMMAND REFERENCE

All ground station commands grouped by subsystem

CORE FLIGHT

```
arm / disarm / emergency_stop / fault_summary / status
cg_status / failure_briefing / soft_stop_status / tick_error_report
```

DISCOVERY AND LEARNING

```
discovery_enable / discovery_disable / discovery_status
seed_hypotheses / hypothesis_results / hypothesis_next
learning_history / mechanism_weight_report / mechanism_weight_reset
persist_checkpoint / persist_load_all / persist_save_all
knowledge_export_now / knowledge_import filename / merge_fleet_knowledge f1 f2
```

MISSION FOCUS

```
focus [name_or_phrase] # or: focus gust_tolerance / focus clear / focus_status
focus_define name=X metric=Y direction=minimize|maximize
```

MAP-ELITES ARCHIVE

```
me_archive_status [verbose=True]
me_best_config V_ms wind_ms intent
me_inject_operator_idea name=X param=val ...
me_archive_import filename / me_archive_export
```

STRATEGY GENOME

```
sg_inject_strategy task=X mech_a=0.6 mech_b=0.4 ...
sg_start_trial task / sg_stop_trial
sg_archive_status [verbose=True] / sg_best_strategy task V wind
sg_breed_all task [method=score_weighted|tournament|mosaic]
sg_breed_report [task]
```

MORPHING HYPOTHESES

```
h33_engage / h33_disengage / h33_status / h33_lift_report
h34_engage / h34_disengage / h35_engage / h35_disengage
h36_engage / h36_disengage / h36_loiter_report / h36_ferry_report
morphing_speed_start / morphing_speed_status / morphing_speed_reset
```

FLAP-GLIDE

```
flap_glide_enable H-FG-1 / H-FG-2 / H-FG-3
flap_glide_disable / flap_glide_status
```

SURFACE CLASSIFIER

```
surface_status / surface_set_water / surface_set_land / surface_auto
surface_set_land_box lat_min lon_min lat_max lon_max
```

WIG FLOOR

```
wig_floor_status / wig_floor_enable / wig_floor_disable
```

STORM RESILIENCE

```
storm_char_report / storm_status
```

AUTOPILOT TRANSIT

```
transit_to lat lon [alt_m] / transit_status / transit_abort
```

GPS

```
gps_status / accept_external_fix lat lon [alt_m] [source]
```

HEALTH AND LOGGING

```
motor_health / motor_peaks / rope_health_status
pitot_status / servo_bandwidth_status / strouhal_status
```

SECTION 3 29 AERODYNAMIC HYPOTHESES

Full list with targets and airspeed ranges

ID	Label	Target	V range
H-ST-1	Strouhal number targeting (optimal)	efficiency	0–28
H-ST-2	Strouhal sprint (high-speed)	speed	9–28
H-ST-3	Strouhal hover (low V endurance)	endurance	0–5
H-PH-1	In-phase cascade (max thrust)	maneuver	0–28
H-PH-2	Anti-phase metachronal (smooth)	smoothness	0–28
H-PH-3	90° phase (standard metachronal)	efficiency	0–28
H-PH-4	180° anti-phase (gust smooth)	smoothness	0–28
H-SK-1	Forward stroke-plane tilt	maneuver	3–28
H-SK-2	Clap-and-fling close approach	efficiency	0–6
H-SK-3	Variable stroke amplitude	efficiency	0–28
H-MO-1	High sweep cruise (75°)	speed	9–28
H-MO-2	Reduced dihedral (0°) cruise	efficiency	9–28
H-MO-3	Ground effect low-altitude hover	endurance	0–4
H-MV-1	Proverse yaw coupling	maneuver	0–28
H-MV-2	Split port/stbd amplitude roll	maneuver	0–28
H-MV-3	Phase-advance pitch authority	maneuver	0–28
H-MV-4	Differential dihedral yaw	maneuver	0–28
H-FG-1	Conservative flap-glide (3s/2s)	range	9–20
H-FG-2	Aggressive flap-glide (2s/3s)	range	9–20
H-FG-3	Dynamic flap-glide (optimizer)	range	9–20
H-SE-1	Surface effect hover over water	endurance	0–4
H-SE-2	Surface effect cruise over water	range	9–18
H-STORM-1	Gust-damping cascade phase	storm_resilience	0–28
H-STORM-2	Asymmetric amplitude gust comp	storm_resilience	0–28
H-STORM-3	Maximum dihedral stability	storm_resilience	0–28
H-STORM-4	Storm headwind penetration	storm_resilience	12–28
H-STORM-5	Dynamic soaring in storm gradient	storm_resilience	14–28
H-STORM-6	Maximum thrust sprint (gust arrest)	storm_resilience	5–28
H-STORM-7	W1 load-shed (3-pair flight)	storm_resilience	0–28

SECTION 4 MISSION FOCI

8 built-in foci with aliases and seeded hypotheses

Name	Scorer measures	Seeds	Persist	Key aliases
gust_tolerance	Att error vs disturbance	H-SK-2,H-MV-3,H-MV-1,H-PH-4	No	windy, gusty, rough air
position_hold	GPS position error	H-MV-3,H-SK-1,H-MV-4	No	drifting, hold still
smooth_flight	IMU RMS + rate	H-PH-4,H-SK-3,H-MO-2	No	smooth, vibration, payload
crosswind_landing	Lateral slip + sink rate	H-MV-1,H-SK-1,H-MV-3	No	crosswind, crabbing
endurance	Metres per watt-second	H-ST-3,H-FG-1/2,H-SE-1	Yes	battery, range, loiter
sprint	Max airspeed in envelope	H-PH-1,H-ST-2,H-MO-1	No	fast, top speed
sar_hover	50% pos + 30% smooth + 20% endurance	H-MV-3,H-PH-4,H-MO-3	Yes	sar, rescue, crew hover
storm_resilience	Stability + margin + position	H-STORM-1..7	Yes	storm, heavy weather, gale

SECTION 5 LEARNING ARCHITECTURE

How discovery, MAP-Elites, and strategy genome interact

Three optimisation systems operate simultaneously, each at a different level of abstraction. They share knowledge through the persist layer.

LEVEL 1 — KINEMATIC DISCOVERY (hill-climber + hypotheses)

discovery_tick runs every tick when enabled. It evaluates the current configuration against the active scorer (Ps or focus score), perturbs parameters, and promotes better configurations. The hypothesis queue injects seeded configurations for structured evaluation. Results feed into the MAP-Elites archive (Level 2).

LEVEL 2 — MAP-ELITES ARCHIVE (125-niche grid)

Each evaluated configuration is placed in the niche matching its measured flight conditions (airspeed bin × wind bin × intent bin). Champions from different niches coexist. Crossover between adjacent niche champions explores the space between known-good configurations. Operator ideas enter via me_inject_operator_idea and are served first ahead of all autonomous candidates. Results inform sg_best_strategy (Level 3).

LEVEL 3 — STRATEGY GENOME (mechanism combination)

sg_evaluate_tick runs during mechanism evaluation sessions. It installs the current genome's mechanism weights (_MW_ROLL, _MW_PITCH, _MW_YAW) and measures task performance directly (roll rate, yaw rate, etc.). Multi-parent crossover combines mechanism strategies from multiple conditions. Operator ideas enter via sg_inject_strategy. Best weights are applied at startup from strategy_genomes.json.

DATA FLOW SUMMARY

From	To	Mechanism
discovery_tick	me_archive	me_tick() called inside discovery_tick def#4
me_archive	strategy_genome	me_cell's kinematic_config attached to genome
mission_focus	discovery_tick	focus_score_tick() replaces Ps scorer
storm_resilience	gust_response_tick	storm_resilience_tick() called from patched GRT
surface_classifier	H-SE-1/2 gating	surface_is_water() called in discovery evaluation
flap_glide	dart_control_tick	flap_glide_tick() called from patched DCT
transit	position_hold_tick	transit_tick() called from patched PHT

SECTION 6 KNOWLEDGE FILES

What persists and where — /home/pi/knowledge/

File	Contents	Written by
mechanism_weights.json	_MW_ROLL, _MW_PITCH, _MW_YAW weights	HQ analyser, mechanism learning
l5_learned.json	L5 yaw blend weights	L5 tick
discovery_promoted.json	Promoted configuration registry	discovery_tick
hypothesis_outcomes.json	H-xx trial results and promotion status	hypothesis_tick
strouhal_log.json	Strouhal number observations by condition	ST scorer
target_rates.json	Learned pitch/roll/yaw rate targets	Mechanism learning
flight_summary.json	Per-flight summary records	persist_save_all
learning_log.jsonl	Append-only event log (all decisions)	All subsystems
me_archive.json	MAP-Elites 125-niche archive	me_tick via discovery_tick
storm_data.json	Wind envelope characterisation across flights	_storm_char_update
strategy_genomes.json	Strategy genome archive (12 tasks)	sg_evaluate_tick
mission_focus.json	Persisted focus state (persist=True foci)	focus_save
morphing_speed_registry.json	H-33/34/35/36 speed leaderboard	Morphing optimizer

All files use atomic writes (tmp→rename) and are version-guarded. The full chain persist_save_sg → persist_save_me → persist_save_storm → persist_save_all (original) runs at shutdown. persist_load_all restores everything at startup.

SECTIONS 7–10 QUICK REFERENCE TABLES

PA constants · GPS levels · Surface classifier · WIG floor

SECTION 7 — PILOT AUTHORITY CONSTANTS

Constant	Value	Meaning
PA_STICK_THRESHOLD	0.05	5% stick → suppress all corrections on that axis
PA_MAX_GUST_CORR_DEG	18.0°	Maximum amplitude correction for gust compensation
PA_MAX_STORM_DIHEDRAL	10.0°	Maximum additive dihedral in storm conditions
PA_CORRECTION_RATE	0.15	Blend rate for adaptive corrections (per tick)

SECTION 8 — GPS DEGRADATION LEVELS

Level	Status	Trigger threshold	Nav mode
0	GPS_OK	HDOP < 2.5 AND ≥ 4 sats	Full GPS
1	GPS_DEGRADED	HDOP > 2.5 OR < 4 sats	Full GPS, warning
2	GPS_LOST	No fix > 3.0 s	Dead reckoning
3	GPS_SPOOFED	Position jump > 50m/tick	Dead reckoning
4	GPS_JAMMED	No fix + RF noise	Dead reckoning

SECTION 9 — SURFACE CLASSIFIER

Evidence	Weight	Threshold for water
GPS coastline (outside CONUS box)	0.90	Alone sufficient (> 0.75 threshold)
IMU vibration below baseline	0.35	Needs + GPS or + wind
Low wind gradient (d(V)/d(h))	0.30	Needs + GPS or + IMU
IWR1443 77GHz radar (optional)	0.95	Alone sufficient — definitive

SECTION 10 — WIG FLOOR SPEED CURVE

Airspeed	Min AGL	Correction at floor
< 5 m/s (inactive)	—	WIG floor off — GEC manages this range
5–9 m/s	interp 0.30–0.50m	AoA boost up to 8°, freq boost up to 0.30Hz
9 m/s	0.50 m	Proportional to floor penetration depth
14 m/s	0.75 m	
20 m/s	1.20 m	
28 m/s (Vne)	1.50 m	Hard limit — surface flight ceiling